$$F = \mu \times W$$

According to the above, the brake efficiency is 100%. However, in practice, 100% of the brake efficiency is considered as not safe and 50% is allowed in the foot operated brake and 30% brake efficiency for hand brakes.

6.12 Stopping distance

Brakes after applying will not stop the vehicles immediately. the vehicle moves certain distance and then only it stops. Normally if the vehicle is running at 30 km/hour the vehicle will stop after 4.5 m when the brake is actuated. This distance is calculated with an approximation. The vehicles stopping distance is varied according to the speed of the vehicle, surface of the road and the type of tread in the tire.

However, in case of emergencies and risky situations, the following factors should be taken into consideration. They are,

i. Time to think for pressing the brake pedal by the driver.
ii. Time taken for pressing brake pedal by the driver.
iii. The time taken for stopping the vehicle after the driver pressed the brake pedal.

6.13 Brake testing

Different methods are used to calculate the brake efficiency. The stop watch test is a simple method among the testing methods.

6.13.1 Stop watch test

For conducting this test the vehicle has to be operated at the speed of 40 km/hr to 50 km/hr. Any symbols in the road can be kept as a mark point. Once if the vehicle is reached that symbol, then the brake pedal has to be pressed and the time has to be noted using the stop watch. The time taken for the vehicle from the mark point and the stopping point of the vehicle should be noted. Distance from the symbol to vehicle get stoped should be noted. From this method we can identify the stopping distance.

6.14 Problems in brake systems, their causes and remedies

Problem 1: Brakes not functioning

S.No	Reasons	Fixing methods
1	If brake shoes were broken or worn out.	Have to be replaced.
2	Poor quality brake shoes	Have to fix good quality brake shoes
3	Oil leakages in the brake shoes	Stop the leakage
4	If the fluid pressure is lower in the hydraulic brake system	Check the fluid leak (or) need to remove the air bubbles.
5	Brake actuating Cable (or) connections become loose	Have to be checked or removed
6	If the brake drum is unevenly worn out.	Has to be corrected for the wear.

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